

RESEARCH ARTICLE

DURAL: Degradation-Resistant Robust Adaptive Localization by LiDAR-Inertial-UWB-Wheel Fusion for Coal Mine Robots

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ABSTRACT

Simultaneous Localization and Mapping (SLAM) in large-scale, complex, global positioning system (GPS)-denied underground coal mines poses significant challenges. In these environments, abnormal conditions hinder sensor performance: GPS unavailability impedes scene reconstruction and geographic referencing; uneven or slippery terrain degrades wheel odometer accuracy; and long, feature-poor tunnels reduce light detection and ranging (LiDAR) effectiveness. To address these challenges, we propose DURAL, a multimodal SLAM framework based on the Iterated Error-State Kalman Filter that fuses multiple sensors from coal mine robots to overcome individual sensor limitations. First, LiDAR-inertial odometry is tightly coupled with Ultra-Wideband (UWB) absolute positioning constraints to establish an absolute coordinate system. Next, the wheel odometer is integrated through tight coupling, enhanced by nonholonomic constraints and vehicle lever arm compensation, to mitigate performance degradation beyond the UWB measurement range. Finally, an adaptive fusion mode switching mechanism dynamically adjusts sensor constraints based on UWB coverage and environmental conditions. Experimental results indicate that our method achieves state-of-the-art accuracy and robustness in both simulated tunnel environments and real-world underground coal mines. In real-world experiments, the system attains an absolute pose error of 0.167 m within the UWB range, maintains a relative pose error of 6.53% outside this range, and improves mapping accuracy to 6.456 cm, significantly outperforming existing approaches in challenging mining scenarios.

1 | Introduction

Simultaneous Localization and Mapping (SLAM) is a fundamental capability for autonomous robots, enabling them to estimate their own states while incrementally constructing a map

of unknown environments using onboard sensors. Due to its significant potential in enhancing autonomous systems, SLAM technology has attracted increasing attention in industrial applications. However, implementing SLAM for underground

Kun Hu and Menggang Li contributed equally to this work and should be considered co-first authors.

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Data Availability Statement

The data that support the findings of this study are available on request from the



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